

STUDENT NUMBER

St Aloysius' College
Year 12 Mid-Year Examinations
2012

EXTENSION 2 MATHEMATICS
(Additional paper)

General Instructions

Reading time – 5 minutes
Working time – 2 hours

- Write using black or blue pen
- Board approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question
- Examination papers must NOT be removed from the examination room.

Total marks - 66

- Attempt all questions

Section I – 6 Marks

- All questions are of equal value
- These are objective response questions.

Section II – 60 Marks

- Questions 7–10 are of equal value.
- Marks for each part are shown in the margin
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Section II Multiple Choice

Select the best response from options A, B, C and D. Give your answers on the multiple choice answer sheet provided.

Question 1

The eccentricity of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 4$ is:

- (A) $\frac{4}{9}$ (B) $\frac{5}{9}$ (C) $\frac{\sqrt{5}}{3}$ (D) $\frac{9}{4}$

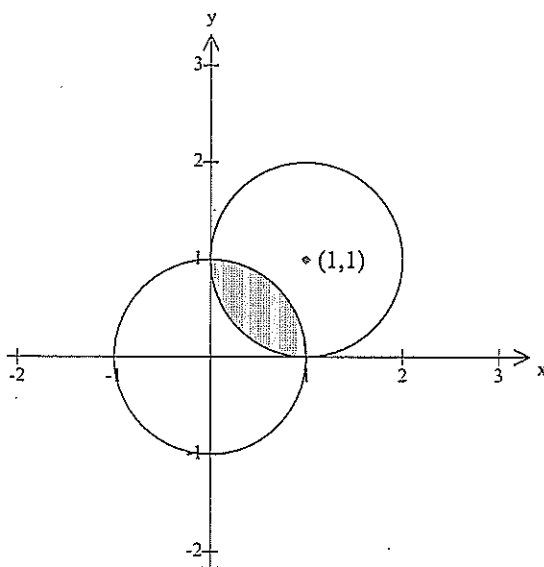
Question 2

Let $z = 3 - i$. What is the value of $\overline{iz}$?

- (A) $-i - 3i$ (B) $-1 + 3i$ (C) $1 - 3i$ (D) $1 + 3i$

Question 3

Consider the Argand diagram below.



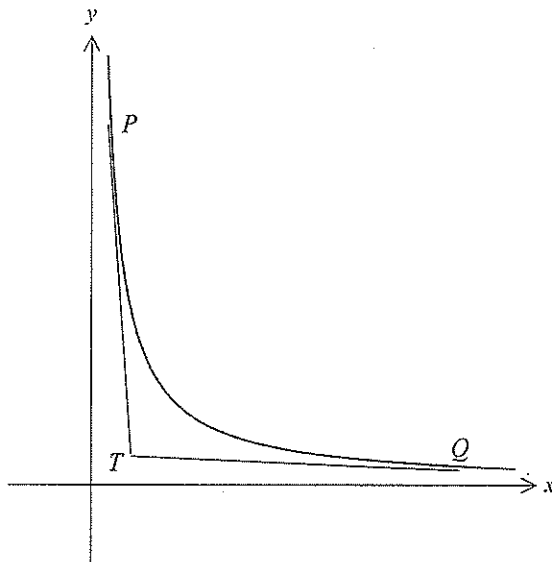
Which inequality could define the shaded area?

- (A) $|z| \leq 1$ and $|z - (1 - i)| \geq 1$
(B) $|z| \leq 1$ and $|z - (1 + i)| \geq 1$
(C) $|z| \leq 1$ and $|z - (1 - i)| \leq 1$
(D) $|z| \leq 1$ and $|z - (1 + i)| \leq 1$

Question 4

The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, $p \neq q$, lie on the same branch of the hyperbola $xy = c^2$.

The tangents at P and Q meet at the point T .



Which of the following expressions is the equation of the tangent to the hyperbola at Q ?

- (A) $x + q^2y = 2cq$
- (B) $x + q^2y = 2c^2$
- (C) $x + p^2y = 2cp$
- (D) $x + p^2y = 2c^2$

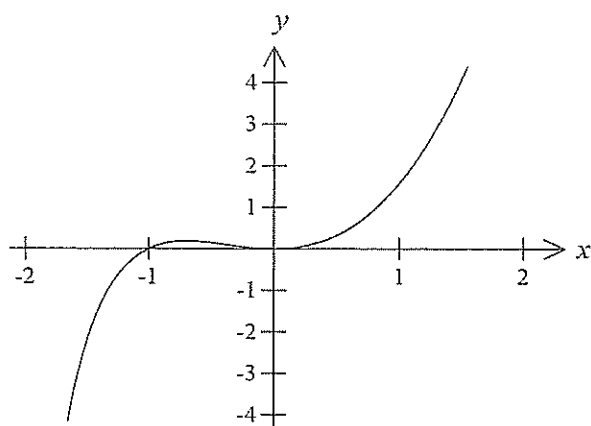
Question 5

What are the solutions of the equation $z^2 = i\bar{z}$?

- (A) 0 and i
- (B) 0 and $-i$
- (C) 0, $-i$, $\frac{\sqrt{3}+i}{2}$ and $\frac{-\sqrt{3}+i}{2}$
- (D) 0, i , $\frac{\sqrt{3}+i}{2}$ and $\frac{-\sqrt{3}+i}{2}$

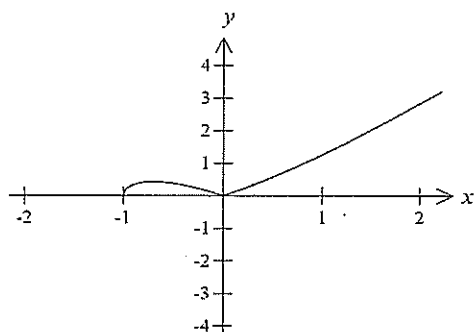
Question 6

The diagram shows the graph of the function $y = f(x)$.

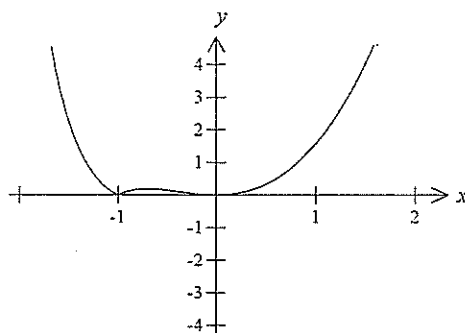


Which of the following is the graph of $y = f(x)^2$?

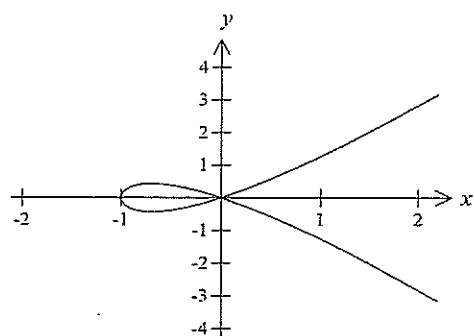
(A)



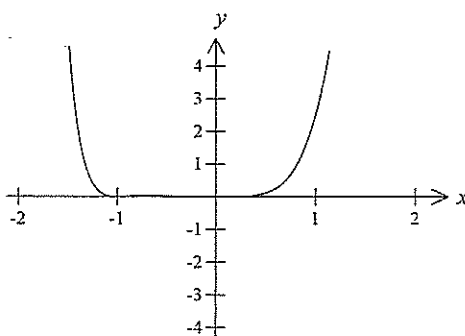
(B)



(C)



(D)



End of Section I

Section II Begin each question in a new answer booklet.
All necessary working is required.

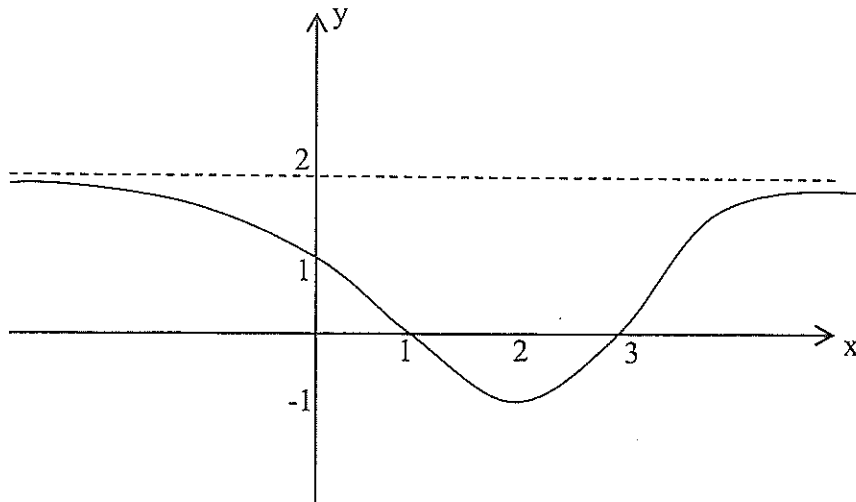
Question 7

- (a) Calculate $|5 - 2i|$ (1)
- (b) For $z = 2 - i$ and $w = -3 + 2i$ find $\frac{w}{z}$ in the form $a + ib$ (2)
- (c) (i) Find $-\sqrt{3} + i$ in mod-arg form (2)
- (ii) Hence, or otherwise, find $(-\sqrt{3} + i)^4$ in the form $a + ib$ (2)
- (d) Find the locus of z which satisfies $\left| \frac{z+1}{z-1} \right| = \sqrt{2}$ and sketch the locus on an Argand diagram. (3)
- (e) Find the roots of $z^3 = \frac{-1 + i\sqrt{3}}{2}$ in mod-arg form. (3)
- (f) If a complex number z has modulus greater than one and a positive acute argument, show on an Argand diagram vectors representing z , $\frac{1}{z}$ and $z - \frac{1}{z}$ (2)

Question 8 **Begin a new booklet**

- (a) Using the domain $-2 \leq x \leq 4$ and a diagram at least half a page in height, (3)
sketch $y = x(x-2)$ and $y^3 = x(x-2)$ on the same diagram.

(b)



Use the above sketch of $y = f(x)$ to draw separate sketches of:

- (i) $y = [f(x)]^2$ (2)
- (ii) $y = \frac{1}{f(x)}$ (2)
- (iii) $y = f(x^2)$ (2)
- (iv) $y = e^{f(x)}$ (2)
- (c) For the relation $x^2y + xy^2 = 16$
- (i) Differentiate implicitly to find y' in terms of x and y . (2)
- (ii) Find the coordinates of any stationary point(s) (2)

Question 9 **Begin a new booklet**

- (a) $P(a \sec \theta, \sqrt{3}a \tan \theta)$ and $Q(a \sec \varphi, \sqrt{3}a \tan \varphi)$ are distinct points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{3a^2} = 1$ such that $PS + QS' = a$ where S and S' are the two foci.

Show that $\sec \theta + \sec \varphi = \frac{1}{2}$ (4)

- (b) For the ellipse E: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola H: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

- (i) State the equations of the asymptotes of H. (1)

- (ii) Show that the point of intersection, P, of an asymptote and E in the first quadrant is $P\left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}}\right)$ (2)

- (iii) Show that the tangent at $Q(a \sec \theta, b \tan \theta)$ on H has the equation (2)

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

- (iv) The tangent at Q passes through P, find the coordinates of Q in exact simplified form. (4)

- (c) For the ellipse and the hyperbola in part (b), find a relationship for the eccentricity of H, e_H , in terms of the eccentricity of E, e_E , when $a^2 \geq b^2$. (2)

Question 10 Begin a new booklet

- (a) (i) For the cubic function $y = x^3 - 3px$ where $p > 0$, find the coordinates of the stationary points in terms of p (2)
- (ii) Deduce that the cubic function $y = x^3 - 3px + q$ where q is real intersects the x axis three times if and only if $q^2 < 4p^3$ (2)
- (b) Suppose that the cubic equation $x^3 - 3px + q = 0$ has the real root α . Further, suppose that there is a real number u such that $\alpha = u + \frac{p}{u}$
- (i) Show by substitution that $u^3 + \frac{p^3}{u^3} + q = 0$ (2)
- (ii) Show that u is real if and only if $q^2 \geq 4p^3$ (2)
- (iii) Using part (a), or otherwise, deduce that if u is real then the cubic equation cannot have three distinct real roots. (2)
- (c) (i) Make a sketch of $\alpha = u + \frac{p}{u}$ in which the u axis is horizontal and the α axis is the vertical. Find and show the turning points. (2)
- (ii) Deduce that, if α is the only distinct real root of the cubic equation in part (b) then $|\alpha| \geq 2\sqrt{p}$ (1)
- (d) Consider again the cubic equation of part (b) with the root $\alpha = u + \frac{p}{u}$ where u is real. Let ω be a complex cube root of unity. Show by substitution that $\beta = u\omega + \frac{p}{u}\bar{\omega}$ is also a root of the cubic equation $x^3 - 3px + q = 0$ (2)

End of Examination

Multiple Choice					
1	2	3	4	5	6
C	C	D	A	C	D

Q27 (a) $|5-2i| = \sqrt{25+4}$
 $= \sqrt{29}$

(1)

(b) $z=2-i; w=-3+2i \therefore \frac{w}{z} = \frac{-3+2i}{2+i}$
 $= \frac{(-3+2i)(2-i)}{4+1}$
 $= \frac{-6+2+i(3+4)}{5}$
 $= \frac{-4}{5} + i \cdot \frac{7}{5}$

(1)

(1)

(c) (i) $|- \sqrt{3}+i| = \sqrt{3+1}$
 $= 2$
 $\arg(-\sqrt{3}+i) = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$ 2nd Quadrant
 $= \frac{5\pi}{6}$ or 150°

$\therefore -\sqrt{3}+i = 2 \operatorname{cis} \frac{5\pi}{6}$ or $2 \operatorname{cis} 150^\circ$

(2)

(ii) $(-\sqrt{3}+i)^4 = 2^4 \operatorname{cis} \frac{10\pi}{3}$
 $= 16 \operatorname{cis} \left(-\frac{2\pi}{3}\right)$
 $= 16 \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$
 $= -8 + 8i\sqrt{3}$

[de Moivre]

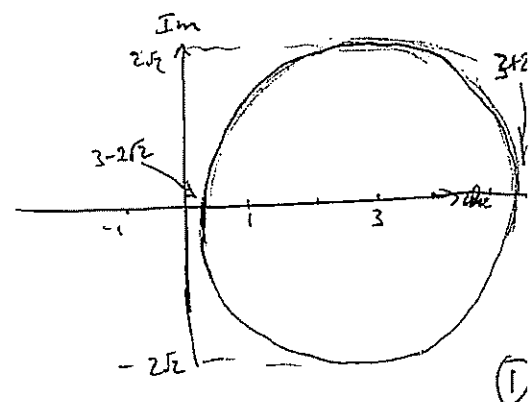
(1)

(1)

(d) $(x+1)^2 + y^2 = 2[(x-1)^2 + y^2]$
 $0 = x^2 - 6x + 1 + y^2$
 $8 = (x-3)^2 + y^2$

(1)

(1)



(1)

(e) $\left|\frac{-1+i\sqrt{3}}{2}\right| = 1 \therefore \text{Let } z = \operatorname{cis} \theta$
 $\therefore \operatorname{cis} 3\theta = \frac{-1+i\sqrt{3}}{2}$

[de Moivre]

(1)

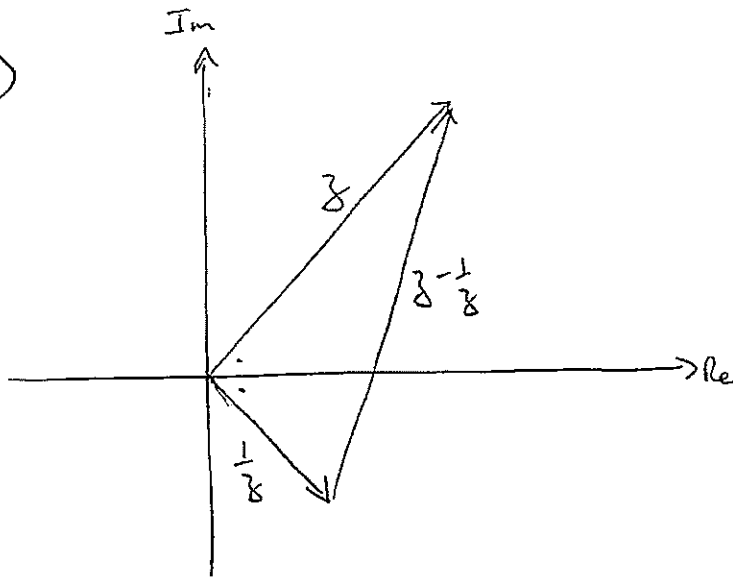
$\cos 3\theta = -\frac{1}{2}$ and $\sin 3\theta = \frac{\sqrt{3}}{2} \therefore 3\theta = \frac{2\pi}{3}, -\frac{4\pi}{3}, \frac{8\pi}{3}$
 $\theta = \frac{2\pi}{9}, -\frac{4\pi}{9}, \frac{8\pi}{9}$

(1)

$\therefore z = \operatorname{cis} \frac{2\pi}{9}, \operatorname{cis} -\frac{4\pi}{9}, \operatorname{cis} \frac{8\pi}{9}$ [or $\operatorname{cis} 40^\circ, \operatorname{cis} -80^\circ, \operatorname{cis} 160^\circ$]

(1)

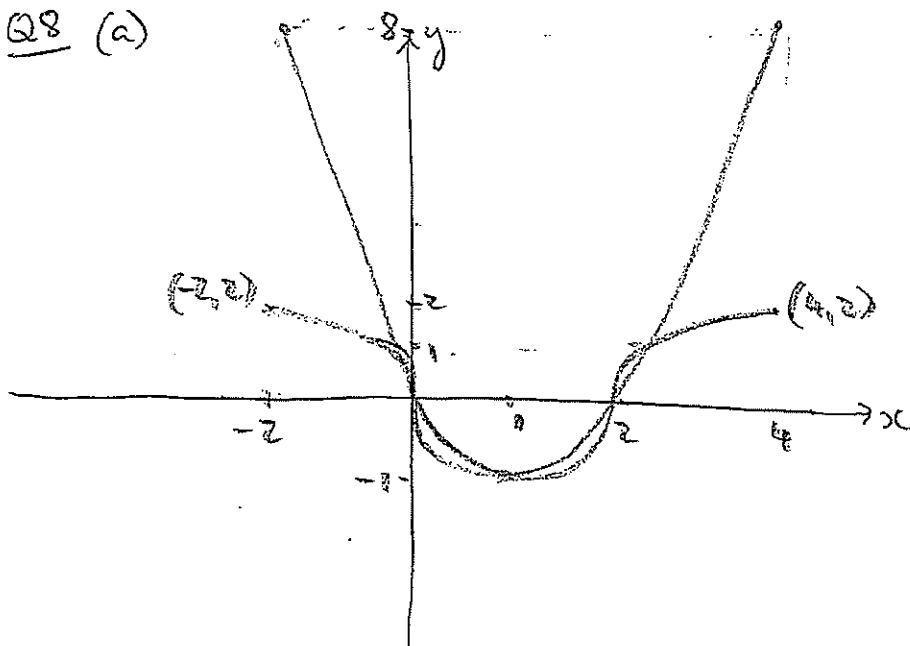
(f)



(1)

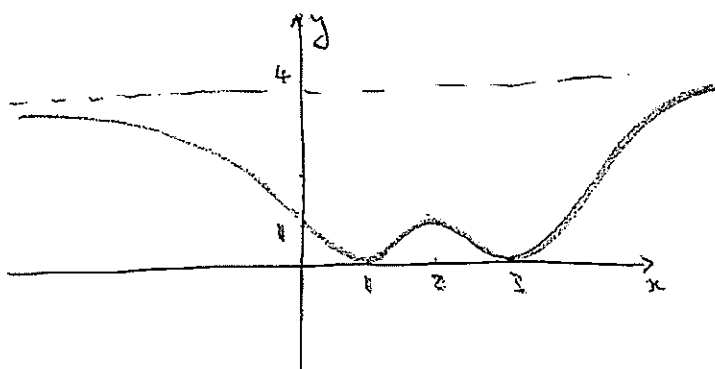
(1)

Q8 (a)

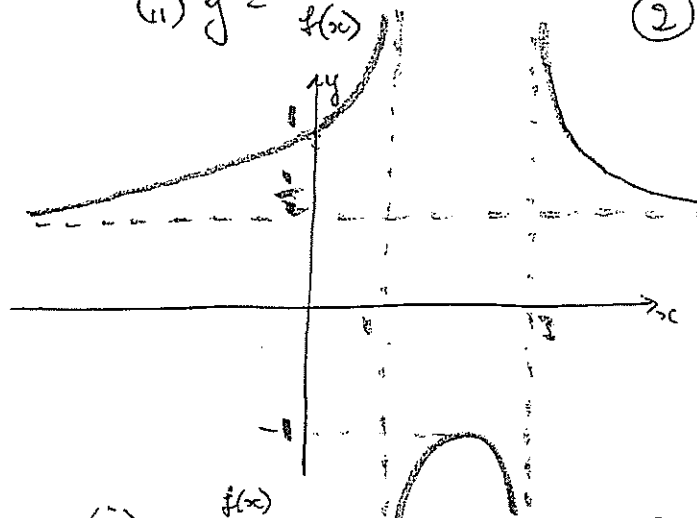


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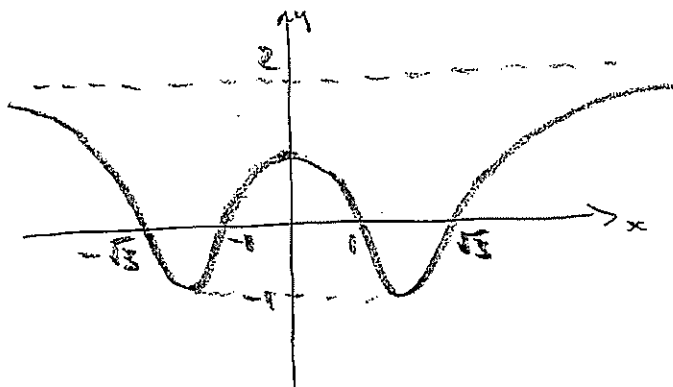
(b) (i) $y = [f(x)]^2$ ②



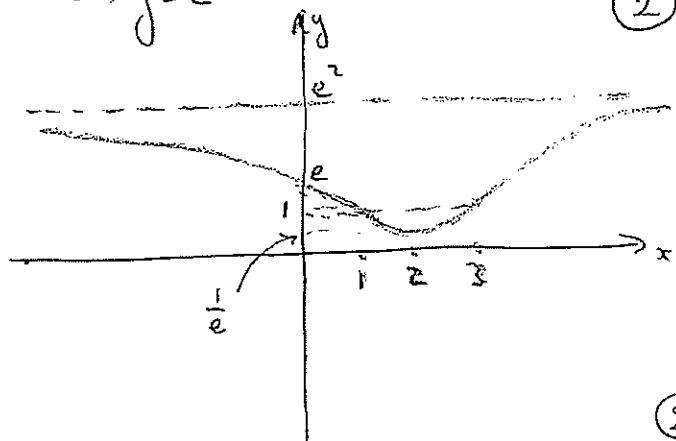
(ii) $y = \frac{1}{f(x)}$ ②



(iii) $y = f(x^2)$ ②



(iv) $y = e^{f(x)}$ ②



(c) (i) $xc^2y + xy^2 = 16 \Rightarrow 2xcy + x^2y' + y^2 + 2xcyy' = 0 \Rightarrow y' = \frac{-y(2x+y)}{x(x+2y)}$

(ii) $y' = 0 \Rightarrow y = -2xc$ only as $y = 0$ is not in the ~~range~~.

$\therefore$ Using $xc^2y + xy^2 = 16 \Rightarrow -2xc^3 + 4xc^3 = 16 \Rightarrow x^3 = 8$

$\therefore$ The only stationary point is $(2, -4)$

②

Q9 (a) $b^2 = a^2(e^2 - 1) \Rightarrow 3a^2 = a^2(e^2 - 1) \Rightarrow e = 2$

$PS + QS' = a$ where $S(2a, 0)$ and $S'(-2a, 0)$

$\therefore 2PN + 2QN' = a$ using $\frac{PS}{PN} = e$; $\frac{QS'}{QN'} = e$

$\therefore \left(a \sec \theta - \frac{a}{2}\right) + \left(a \sec \phi + \frac{a}{2}\right) = \frac{a}{2}$

$\therefore \sec \theta + \sec \phi = \frac{1}{2}$

(b) (i) $y = \pm \frac{b \sec \theta}{a}$

(ii) For P, $\frac{x^2}{a^2} + \frac{b^2 c^2}{a^2 b^2} = 1 \Rightarrow x = \pm \frac{a}{\sqrt{2}}$

$\therefore y = \pm \frac{b}{\sqrt{2}}$

P is in the 1st Quadrant $\therefore P\left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}}\right)$

(iii) $y' = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$
 $= \frac{b \sec^2 \theta}{a \sec \theta \tan \theta}$
 $= \frac{b \sec \theta}{a \tan \theta}$

$\therefore$ Tangent at Q

$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$

$\sec^2 \theta - \tan^2 \theta = \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b}$

$\therefore 1 = \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b}$

(iv) Substituting P $\therefore 1 = \frac{\sec \theta}{\sqrt{2}} - \frac{\tan \theta}{\sqrt{2}}$

$\therefore \sqrt{2} + \tan \theta = \sec \theta$

$2 + 2\sqrt{2} \tan \theta + \tan^2 \theta = \tan^2 \theta + 1$

$\therefore \tan \theta = -\frac{1}{2\sqrt{2}}$

$\therefore \sec \theta = \sqrt{2} - \frac{1}{2\sqrt{2}}$
 $= \frac{3}{2\sqrt{2}}$

$\therefore Q\left(\frac{3a}{2\sqrt{2}}, -\frac{b}{2\sqrt{2}}\right)$

(c) $b^2 = a^2(1 - e_e^2)$ and $b^2 = a^2(e_h^2 - 1)$

$\therefore e_h^2 - 1 = 1 - e_e^2$

$e_h = \sqrt{2 - e_e^2}$

Q10 (a) (i) $y = x^3 - 3px$; $p > 0$

$$y' = 3x^2 - 3p \quad \therefore y' = 0 \Rightarrow x = \pm\sqrt{p}$$

$\therefore (\sqrt{p}, -2p\sqrt{p})$ and $(-\sqrt{p}, 2p\sqrt{p})$ are stationary points

(ii) $y = x^3 - 3px + q$ is the function $y = x^3 - 3px$ shifted vertically by the distance $|q|$. The function will continue to have 3 distinct x intercepts if and only if that shift is less than the height of the stationary points i.e. $-2p\sqrt{p} < q < 2p\sqrt{p} \Rightarrow q^2 < 4p^3$

OR The stationary points of $y = x^3 - 3px + q$ are at $(\sqrt{p}, -2p\sqrt{p} + q)$ and $(-\sqrt{p}, 2p\sqrt{p} + q)$

There will be 3 distinct x intercepts iff these points are on opposite sides of the x axis

$$\Rightarrow q < 2p\sqrt{p} \text{ if } q > 0 \text{ or } q > -2p\sqrt{p} \text{ if } q < 0$$

$$\therefore q^2 < 4p^3 \text{ for all } q.$$

(b)(i) Since α is a root $\alpha^3 - 3p\alpha + q = 0$

$$\text{If } \alpha = u + \frac{p}{u} \text{ then } \left(u + \frac{p}{u}\right)^3 - 3p\left(u + \frac{p}{u}\right) + q = 0$$

$$\therefore u^3 + 3pu + \frac{3p^2}{u} + \frac{p^3}{u^3} - 3pu - \frac{3p^2}{u} + q = 0$$

$$\therefore u^3 + \frac{p^3}{u^3} + q = 0 \quad (1)$$

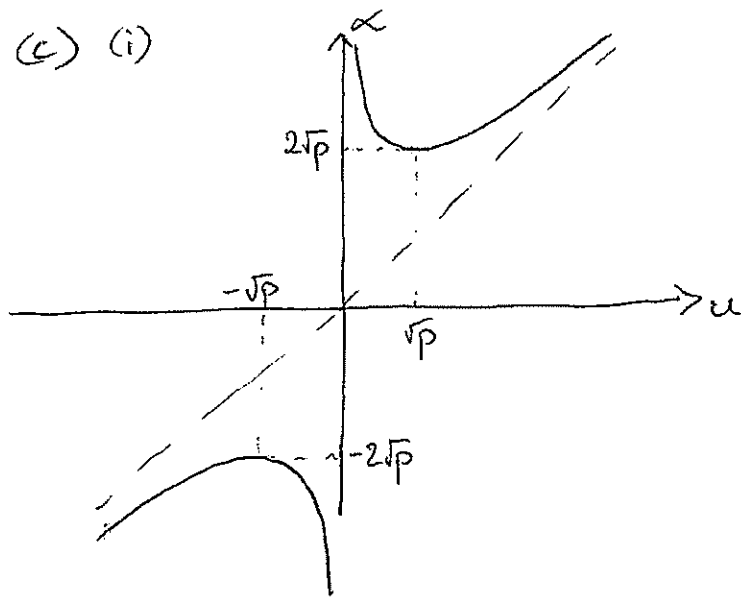
$$(ii) (1) \Rightarrow u^6 + qu^3 + p^3 = 0 \Rightarrow \Delta = q^2 - 4p^3$$

u^3 , and hence u , is real iff $\Delta \geq 0$

$$\therefore q^2 \geq 4p^3$$

(iii) This set of values for q for which u is real is the exact opposite of the set of values in part (a) (ii) for which there are 3 distinct real ~~roots~~ x intercepts, and hence roots. Hence u being real and there being 3 distinct real roots are mutually exclusive.

(c) (i)



$$\alpha = u + \frac{p}{u}$$

$$\frac{d\alpha}{du} = 0 \Rightarrow u^2 = p$$

$$u = \pm \sqrt{p}$$

2

(ii) Since all cubic functions have at least one real root, if u is real then $\alpha = u + \frac{p}{u}$ is the only real root.

The above graph shows the values of α for all real $u \neq 0$.

$$\therefore \alpha \geq 2\sqrt{p} \text{ or } \alpha \leq -2\sqrt{p}$$

$$\Rightarrow |\alpha| \geq 2\sqrt{p}$$

(d) Substituting $x = \beta$ into $x^3 - 3px + q$

$$\beta^3 - 3p\beta + q = \left(u\omega + \frac{p}{u}\bar{\omega}\right)^3 - 3p\left(u\omega + \frac{p}{u}\bar{\omega}\right) + q$$

$$= u^3\omega^3 + 3pu\omega^2\bar{\omega} + \frac{3p^2}{u}\omega\bar{\omega}^2 + \frac{p^3}{u^3}\bar{\omega}^3 - 3pu\omega - \frac{3p^2}{u}\bar{\omega} + q$$

Now $\omega^3 = \bar{\omega}^3 = 1$; $\omega\bar{\omega} = 1 \therefore \omega^2\bar{\omega} = \omega$ and $\omega\bar{\omega}^2 = \bar{\omega}$

$$\therefore \beta^3 - 3p\beta + q = u^3 + \cancel{3pu\omega} + \cancel{\frac{3p^2}{u}\bar{\omega}} + \frac{p^3}{u^3} - \cancel{3pu\omega} - \cancel{\frac{3p^2}{u}\bar{\omega}} + q$$

$$= u^3 + \frac{p^3}{u^3} + q$$

$$= 0 \quad \text{from part (b)(i)}$$

$\therefore \beta$ is also a root of the equation.